

591 (continued).

A SMITH'S PRIZE PAPER AND DISSERTATION.

[Continued, pp. 231–236 of Vol. IX.]

viz. since $X + Y = -P$, and $XY = Q$, this is

$$(P^2 - 4Q)X'^2Y'^2 = 0,$$

and writing also

$$X = -\frac{1}{2}\{P + \sqrt{(P^2 - 4Q)}\}, \quad Y = -\frac{1}{2}\{P - \sqrt{(P^2 - 4Q)}\},$$

we find

$$X'Y' = \frac{1}{4} \left\{ P'^2 - \frac{(PP' - 2QQ')^2}{P^2 - 4Q} \right\};$$

the differential equation thus is

$$(P^2 - 4Q) \left\{ P'^2 - \frac{(PP' - 2QQ')^2}{P^2 - 4Q} \right\} = 0.$$

The application to the theory of singular solutions is that, in the case where the function $(1, P, Q, \dots)(c, 1)^n$ breaks up into rational factors $c - X, c - Y, \dots$, the factor $\zeta = (X - Y)^2(X - Z)^2 \dots$ divides out and should be rejected from the differential equation, which in its true form is $X'Y'Z' \dots = 0$; viz. this is what we obtain immediately, considering the given integral equation as meaning the system of curves $c - X = 0, c - Y = 0, \dots$, and there is not really any singular solution; whereas in the case where the factors are not rational, the factor in question, when the product $X'Y'Z' \dots$ is expressed in terms of the coefficients $P, Q, \dots$, and their derived coefficients does not divide out from the equation; and in this case, equated to zero, it gives a proper singular solution of the equation.

11. In the theory of elliptic motion, v denoting the mean anomaly and e the eccentricity, if m' be an angle such that $\tan \frac{1}{2}v = \sqrt{\frac{1+e}{1-e}} \tan \frac{1}{2}m'$, find in terms of e, m' the mean anomaly m .

Taking as usual u for the eccentric anomaly, to commence the solution write down

$$\tan \frac{1}{2}u = \sqrt{\frac{1+e}{1-e}} \frac{1 - \tan \frac{1}{2}v}{1 + \tan \frac{1}{2}v} \cdot \tan \frac{1}{2}v,$$

that is,

$$\tan \frac{1}{2}u = \sqrt{\frac{1-e^2}{(1-e)^2}} \tan \frac{1}{2}m',$$

and u being given hereby as a function of m' , we have by substitution in the equation $m = u - e \sin u$, to find m as a function of m' .

A creditable approximate solution would be $m = m' + 0 \cdot e$, viz. this would be to show that neglecting terms in e^2 , &c., we have $m = m'$. In fact, taking e small, we have

$$\tan \frac{1}{2}u = (1 + e) \tan \frac{1}{2}m',$$

and thence if $m = m' + x$, we have

$$\tan \frac{1}{2}m' + \frac{1}{2}x \sec^2 \frac{1}{2}m' = (1 + e) \tan \frac{1}{2}m';$$

that is,

$$x = 2e \cos^2 \frac{1}{2}m' \tan \frac{1}{2}m' = e \sin m'; \quad m = m' + e \sin m',$$

and

$$m = m' + e \sin m' - e \sin(m' + \dots) = m' + 0 \cdot e.$$

The complete solution would be obtained by expanding u in terms of e, m' from the equation $\tan \frac{1}{2}u = \sqrt{\frac{1+e}{1-e}} \tan \frac{1}{2}m'$ (which is of the form $\tan \frac{1}{2}u = n \tan \frac{1}{2}m'$, giving for u a known series $= m' + \text{multiple sines of } m'$), and then observing that the same equation leads to

$$\sin u = \frac{\sqrt{(1-e^2)} \sin m'}{1 - e \cos m'},$$

we have

$$m = \text{series} - e \frac{\sqrt{(1-e^2)} \sin m'}{1 - e \cos m'},$$

where the second term has also to be expanded in a series of multiple sines of m' , which can be done without difficulty.

12. If (u, v) are given functions of the coordinates (x, y) , neither of them a maximum or a minimum at a given point O ; and if through O we draw Ox' in the direction in which v is constant and u increases, and Oy' in the direction in which u is constant and v increases; then the rotation (through an angle not greater than π), from Ox' to Oy' is in the same direction with that from Ox to Oy , or in the contrary direction, according as $\frac{du}{dx} \frac{dv}{dy} - \frac{du}{dy} \frac{dv}{dx}$ is positive or negative.

The theorem has not, so far as I am aware, been noticed, and it seems to be one of some importance; there is no difficulty in it, but the answer requires some care in writing out; of course where the whole question is one of sign and direction, the omission to state that a subsidiary quantity is positive may render an answer worthless.

It depends on the following lemma: Consider the triangle $OX'Y'$, such that Ox, Oy being any rectangular axes through the origin O , the coordinates of X' are h, k , and those of Y' are h_1, k_1 ; then considering the area as positive, the double area is $= \pm(hk_1 - h_1k)$, viz. the sign is $+$ or $-$ according as the rotation from OX' to OY' (through an angle less than π) is in the same direction with that from Ox to Oy , or in the contrary direction; or, what is the same thing, $hk_1 - h_1k$ is in the first case positive and in the second case negative.

To show this, suppose for a moment that the lines OX', OY' are each of them in the quadrant xOy , say in the first quadrant, the inclination of OY' to Ox exceeding that of OX' to Ox ; then h, k, h_1, k_1 are all positive, and $\frac{k_1}{h_1} > \frac{k}{h}$, that is, $hk_1 - h_1k$ is $+$, and the rotation from OX' to OY' is in the same direction as that from Ox to Oy ; or the lemma holds good. Now OX' remaining fixed, let OY' revolve in the direction Ox to Oy ; so long as OY' remains in the first quadrant, $\frac{k_1}{h_1}$ continues to increase, and we have always $\frac{k_1}{h_1} > \frac{k}{h}$, and $hk_1 - h_1k = +$; when OY' comes into the second quadrant (h, k being always positive), h_1 is negative and k_1 positive, consequently $hk_1 - h_1k$ is the sum of two positive terms, and therefore $= +$; as OY' continues to revolve and passes into the third quadrant, we have h_1, k_1 each negative, but $\frac{k_1}{h_1} < \frac{k}{h}$, and therefore $hk_1 - h_1k$ still $= +$; when, however, OY' comes into the position opposite to OX' , then $\frac{k_1}{h_1} = \frac{k}{h}$, and $hk_1 - h_1k$ is $= 0$; and when OY' , continuing in the third quadrant, has passed the position in question, we have $\frac{k_1}{h_1} > \frac{k}{h}$, and therefore $hk_1 - h_1k = -$, but now the angle $X'OY'$ measured in the original direction has become $> \pi$, and the rotation OX' to OY' through an angle less than π will be in the opposite direction, that is, in the direction opposite to that from Ox to Oy ; and, similarly, when OY' passes into the fourth quadrant, and until, passing into the first quadrant, it approaches the position OX' , the sign of $hk_1 - h_1k$ will be $-$, and the rotation will be in the direction contrary to that from Ox to Oy . The lemma is thus true for any position of OX' in the first quadrant; and the like reasoning would show that it is true for any position of OX' in the second, the third, or the fourth quadrant; hence the lemma is true generally.

This being so, taking a new origin, let the coordinates of O be x, y ; and drawing through O the axes Ox', Oy' as directed, let X' be the point belonging to the values $u + \delta u, v$ of (u, v) , and Y' the point belonging to the values $u, v + \delta v$ of (u, v) ; taking δu positive, X' will be on Ox' in the direction O to x' , and similarly taking δv positive, Y' will be on Oy' in the direction O to y' . Taking as before (h, k) for the

coordinates of X' , and (h_1, k_1) for the coordinates of Y' , these coordinates being measured from the point O as origin, we have

$$\delta h = \frac{du}{dx} h + \frac{du}{dy} k,$$

$$\delta k = \frac{dv}{dx} h + \frac{dv}{dy} k;$$

whence, writing for a moment $J = \frac{du}{dx} \frac{dv}{dy} - \frac{du}{dy} \frac{dv}{dx}$, we have

$$Jh = +\frac{dv}{dy} \delta u, \quad Jk = -\frac{dv}{dx} \delta u.$$

And in like manner

$$\delta h = \frac{du}{dx} h_1 + \frac{du}{dy} k_1,$$

$$\delta k = \frac{dv}{dx} h_1 + \frac{dv}{dy} k_1;$$

whence

$$Jk_1 = -\frac{du}{dy} \delta v, \quad Jh_1 = \frac{du}{dx} \delta v;$$

and hence

$$J^2(hk_1 - h_1k) = \left(\frac{du}{dx} \frac{dv}{dy} - \frac{du}{dy} \frac{dv}{dx} \right) J \delta u \delta v = J \delta u \delta v,$$

that is,

$$hk_1 - h_1k = \frac{1}{J} \delta u \delta v,$$

and $\delta u, \delta v$ being as above each of them positive, J has the same sign as $hk_1 - h_1k$. But the rotation from OX' to OY' is in the same direction as that from Ox to Oy , or in the contrary direction, according as $hk_1 - h_1k$ is + or −, that is, according as $J = \frac{du}{dx} \frac{dv}{dy} - \frac{du}{dy} \frac{dv}{dx}$ is + or −; which is the theorem in question.

13. Write a dissertation on:

The theory and constructions of Perspective.

In Perspective we represent an object in space by means of its central projection upon a plane: viz. any point P_1 ¹ of the object is represented by P' , the intersection with the plane of projection of the line D_1P_1 from the centre of projection (or say the eye) D_1 to the point P_1 ; and considering any line or curve in the object, this is represented by the line or curve which is the locus of the points P' , the projections of the corresponding points P_1 of the line or the curve in the object.

The fundamental construction in perspective is derived from the following considerations: viz. considering through P_1 (fig. 1) a line meeting the plane of projection in Q , and drawing parallel thereto through D_1 a line to meet the plane of projection in M and joining the points M, Q , then the lines D_1M, MQ, QP_1 are in a plane; that is, the plane through D_1 and the line P_1Q meets the plane of projection in MQ ;

[Figure 1: A schematic showing the eye D_1 outside the plane of projection, a point P_1 of the object and the projection line through D_1 to P_1 meeting the plane in P' . An auxiliary line through P_1 meets the plane in Q , with a parallel line through D_1 meeting the plane in M ; the joined segment MQ shows the trace of the plane through D_1P_1Q .]

and consequently the projection P' of any point P_1 in the line P_1Q lies in the line QM ; and not only so, but considering only the points P_1 of this line which lie behind the plane of projection (D_1 being considered as in front of it), the projections lie on the terminated line MQ ; viz. Q is the projection of the point Q , and M the projection of the point at infinity on the line QP_1 ; or, if we please, the finite line QM is the projection of the line $QP_1\infty$.

¹The subscript unity is used to denote a point not in the plane of projection, considered as a point out of this plane; a point in the plane of projection, used in the constructions of perspective as a conventional representation of a point P_1 , will be denoted by the same letter P without the subscript unity. And the like as regards D_1 and D .

If we consider a set of lines parallel to P_1Q , these all give rise to the same point M , and thus their projections MQ all pass through this point M , which is said to be the “vanishing point” of the system of parallel lines. Again, if we consider any two or more lines through P_1 , to each of these there correspond different points M and Q , and, therefore, a different line MQ , but these all intersect in a common point P' which is the projection of P_1 . If the lines are all in one and the same plane through P_1 , then the locus of the points Q is a line, the intersection of this plane with the plane of projection, say the “trace” line; and the locus of the points M is a parallel line, the intersection of the parallel plane through D_1 with the plane of projection; say this is the “vanishing line” for the plane in question.

A construction in perspective presupposes a conventional representation on the plane of projection (or say on the paper) as well of the position of the eye as of the object to be projected. If for simplicity we suppose the object to be a figure in one plane, then this plane intersects the paper in a trace line, and we may imagine the plane made to rotate about the trace line until it comes to coincide with the paper, and we have thus the plane object conventionally represented on the paper. Similarly considering the parallel plane through the eye D_1 , and regarding D_1 as a point of this plane, the plane meets the paper in the vanishing line, and we may imagine the plane made to rotate (in the direction opposite to that of the first rotation) until it comes to coincide with the paper, bringing the point D_1 to coincide with a point D of the paper. We have thus the “point of distance” D , being a conventional representation on the paper of the position of the eye D_1 ; but which point D has, observe, a different position for different directions of the plane of the object.

To fix the ideas, suppose the plane of projection to be vertical, and the plane of the object to be a horizontal plane situate below the eye. The trace line will be represented by a horizontal line HH' (fig. 2), and the object by a figure in the plane of the paper below the line HH' such that, bending this portion of the paper backwards through a right angle round HH' , the figure would be brought to coincide with the object². The vanishing line will be a horizontal line KK' above HH' , and the eye will be represented by a point D above KK' , in suchwise that, bending the upper part of the paper round KK' forwards through a right angle, the point D would come to coincide with the position D_1 of the eye. This being so, taking any line PQ in the representation of the object, we draw through D the parallel line DM , and then joining the points M and Q , we have MQ as the perspective representation of the line $QP\infty$, which represents a line QP_1 of the object. And drawing through P any number of lines, each of these gives a point Q and a point M , but the lines MQ all meet in a common point P' , which is the perspective representation of the point P ; which point P' may, it is clear, be obtained as the intersection with any one line MQ of the line DP drawn to join P with the point of distance D . The plane of the object has for convenience been taken to be horizontal; but its position may be any whatever, and in particular the construction is equally applicable in the case where the plane is vertical.

[Figure 2: A vertical plane of projection. A horizontal trace line HH' , above it a parallel horizontal vanishing line KK' , and above KK' the eye-point D with its foot S on KK' . Below HH' in the same paper plane lies the rotated horizontal object plane, containing a line PQ . Joining D to P and drawing $DM \parallel PQ$ to meet KK' in M , then MQ is the perspective of the line $QP\infty$, and $P' = DP \cdot MQ$ is the perspective of P .]

In the case of an object not in one plane, any point Q_1 of the object may be determined by means of its projection by a vertical line upon a given horizontal plane, say this is P_1 , and of its altitude Q_1P_1 above this plane. We in fact determine the object by means of its groundplan, and of the altitudes of the several points thereof. It is easy, from the foregoing principles, to see that, drawing through P the vertical line PQ equal to the altitude, and joining the points Q, D , then the vertical line through P' meets this line QD in a point Q' , which will be the perspective representation of Q_1 . We have thus a construction applicable to any solid figure whatever.

²It is assumed in the text, that the figure on the paper is equal in magnitude to the object; but practically the figure is drawn on a reduced scale, the distance between the lines KK', HH' , and the distance DS (representing respectively the distance between the parallel planes, and the distance of the eye from the plane of projection) being drawn on the same reduced scale.

[From the *Messenger of Mathematics*, vol. iv. (1875), pp. 17—20.]

In a note “On the Mercator’s-projection of a surface of revolution” read before the British Association, [555, (5)], I remarked that the surface might be, by its meridians and parallels, divided into infinitesimal squares; and that these would be on the map represented by two systems of parallel lines at right angles to each other, dividing the map into infinitesimal squares; and that, by taking the squares not infinitesimal but small, for instance, by considering the meridians at intervals of 10° or 5° , we might approximately construct a Mercator’s-projection of the surface. But it is worth while, for the skew hyperboloid of revolution, to develop analytically the ordinary accurate solution.

Taking the equation of the surface to be

$$\frac{x^2 + y^2}{a^2} - \frac{z^2}{c^2} = 1,$$

(or, if as usual $a^2 + c^2 = a^2 e^2$, then $x^2 + y^2 - (e^2 - 1)z^2 = a^2$), and writing $x = r \cos \theta$, $y = r \sin \theta$, the meridians corresponding to the several longitudes θ are in the map represented by the parallel lines $X = a\theta$, and the parallels corresponding to the several values of z are in the map represented by a set of parallel lines $Z = f(z)$, the form of the function being so determined that the infinitesimal rectangles on the map are similar to those on the surface. The required relation is readily found to be

$$Z = \int \frac{\sqrt{\{(a^2 + c^2)z^2 + c^4\}}}{z^2 + c^2} dz,$$

where the integral is taken from the value $z = 0$.

The substitution which first presents itself is to write herein $z = \frac{c^2}{\sqrt{(a^2 + c^2)}} \tan \phi$; or, what is the same thing,

$$z = a\left(e - \frac{1}{e}\right) \tan \phi,$$

where observe that $a\left(e - \frac{1}{e}\right)$ is the distance between a focus and its corresponding directrix. The equation of the surface is satisfied by writing therein $\sqrt{(x^2 + y^2)} = a \sec \psi$, $z = c \tan \psi$, and ψ as thus defined is the “parametric latitude”; hence the foregoing angle ϕ is a deduced latitude connected with the parametric latitude ψ by the equation

$$\tan \phi = \frac{c}{a\left(e - \frac{1}{e}\right)} \tan \psi, \quad = \frac{c}{\sqrt{(c^2 - 1)}} \tan \psi.$$

The resulting formula in terms of ϕ is

$$Z = 2c^2 \int_0^\phi \frac{(1 + \zeta^2)^2 d\zeta}{c^2(1 + \zeta^2)^2 + a^2(1 - \zeta^2)^2},$$

or, if we write herein $\zeta = \tan \frac{1}{2}\phi$, the formula becomes

$$Z = \int \frac{d\phi}{\cos \phi (c^2 + a^2 \cos^2 \phi)},$$

viz. the function under the integral sign is rational. The expression is, however, complicated, and a more simple formula is obtained by using instead of ϕ the parametric latitude ψ ; viz. we have $z = c \tan \psi$, and thence

$$Z = \int \frac{\sqrt{(a^2 \sin^2 \psi + c^2)}}{\cos \psi} d\psi,$$

or, putting herein

$$\sin \psi = \frac{c}{a} \cdot \frac{u}{\sqrt{(1 - u^2)}},$$

and therefore

$$\cos^2 \psi = \frac{a^2 - (a^2 + c^2) u^2}{a^2(1 - u^2)},$$

and

$$a^2 \sin^2 \psi + c^2 = \frac{c^2}{1 - u^2}, \quad \cos \psi d\psi = \frac{c}{a} \cdot \frac{du}{(1 - u^2)^{\frac{3}{2}}},$$

the formula becomes

$$Z = c^2 a \int_0^u \frac{du}{(1 - u^2) \{a^2 - (a^2 + c^2) u^2\}},$$

or, what is the same thing,

$$Z = c^2 a \int_0^u \frac{du}{(1 - u^2)(1 - e^2 u^2)},$$

viz. we thus have

$$Z = \frac{1}{2} a \left\{ \log \frac{1 + u}{1 - u} - e \log \frac{1 - eu}{1 + eu} \right\},$$

the logarithms being hyperbolic.

As already mentioned, u is connected with the parametric latitude ψ by the equation

$$\sin \psi = \frac{c}{a} \cdot \frac{u}{\sqrt{(1 - u^2)}}, \quad = \frac{u \sqrt{(e^2 - 1)}}{\sqrt{(1 - u^2)}},$$

that is,

$$\sin \psi = \sqrt{(e^2 - 1)} \tan p, \quad \text{if } u = \sin p,$$

or conversely

$$u = \frac{\sin \psi}{\sqrt{(e^2 - 1 + \sin^2 \psi)}},$$

so that the point passing to infinity along the branch of the hyperbola, or ψ passing from 0 to 90° , u passes from 0 to $\frac{1}{e}$; and for $u = \frac{1}{e}$ the value of Z becomes, as it should do, infinite. The value of z in terms of u is

$$z = \frac{\sqrt{(e^2 - 1)} u}{\sqrt{(1 - e^2 u^2)}}, \quad \text{or conversely} \quad u = \frac{z}{\sqrt{(c^2 + e^2 z^2)}},$$

and we have, moreover,

$$u = \frac{1}{e} \cdot \frac{e\zeta}{1 + \zeta^2}, \quad = \frac{1}{e} \sin \phi, \quad = (\text{as before}) \frac{\sin \psi}{\sqrt{(e^2 - 1 + \sin^2 \psi)}}.$$

It will be recollected that, in the Mercator's-projection of the sphere, the longitude and latitude being θ, ϕ , the values of X, Z are

$$X = a\theta, \quad Z = \log \tan\left(\frac{\pi}{4} + \frac{1}{2}\phi\right),$$

the logarithm being hyperbolic.

In the case of the rectangular hyperbola $a = c, = 1$ suppose,

$$e = \sqrt{2}, \quad z = \tan \psi, \quad u = \frac{\sin \psi}{\sqrt{(1 + \sin^2 \psi)}}, \quad = \sin p, \quad \text{if } \sin p = \tan \psi;$$

whence

$$Z = \frac{1}{2} h \cdot l \frac{\tan(45^\circ - \frac{1}{2}p)}{\tan(45^\circ + \frac{1}{2}p)} - \frac{1}{2} \sqrt{(2)} h \cdot l \frac{\tan(22^\circ 30' - \frac{1}{2}p)}{\tan(22^\circ 30' + \frac{1}{2}p)},$$

the first term being of course

$$= h \cdot l \tan(45^\circ - \frac{1}{2}p), \quad \text{or } = -h \cdot l \tan(45^\circ + \frac{1}{2}p).$$

Transforming to ordinary logarithms, this is

$$Z = \frac{1}{\sqrt{(2)} \log e} \left[-\sqrt{(2)} \log \tan(45^\circ + \frac{1}{2}p) + \{ \log \tan(22^\circ 30' + \frac{1}{2}p) - \log \tan(22^\circ 30' - \frac{1}{2}p) \} \right],$$

say this is

$$Z = \frac{1}{\sqrt{(2)} \log e} (-A + B),$$

where

$$\begin{aligned} A &= \sqrt{(2)} \log \tan(45^\circ + \tfrac{1}{2}p), \\ B &= \log \tan(22^\circ 30' + \tfrac{1}{2}p) - \log \tan(22^\circ 30' - \tfrac{1}{2}p). \end{aligned}$$

Taking ψ as the argument, I tabulate $z, = \tan \psi$, and $Z \cdot \sqrt{(2)} \log e, = -A + B$, as shown in the annexed table: the last column of which gives, therefore, the positions of the several parallels of $5^\circ, 10^\circ, \dots, 85^\circ$; the interval of 5° between two meridians is, on the same scale,

$$\frac{\sqrt{(2)} \log e}{180} \cdot \pi = (1.4142136)(.4342945)(.0872665) = .05360,$$

viz. this is nearly equal to the arc of meridian 0° to 5° , and the table shows that the arcs $0^\circ - 5^\circ, 5^\circ - 10^\circ$, &c. continually increase as in a Mercator's-projection of the sphere, but more rapidly; there is, however, nothing in this comparison, since the determination of latitude on the hyperboloid by the equation $z = \tan \psi$ is altogether arbitrary.

ψ	$z = \tan \psi$	p	A	B	$-A + B$
0°	0.	$0^\circ 0' 0''$	0.	0.	0.
5	.08749	$4^\circ 58' 51''$	.05348	.10719	.05370
10	.17632	$9\ 51\ 0$	.10611	.21439	.10827
15	.26795	$14\ 30\ 40$	.15724	.32174	.16450
20	.36397	$18\ 53\ 0$	.20619	.42943	.22324
25	.46631	$22\ 54\ 30$	.25342	.53780	.28539
30	.57735	$26\ 34\ 0$	.29557	.64758	.35201
35	.70020	$29\ 50\ 20$	.33538	.75959	.42421
40	.83910	$32\ 44\ 0$	.37170	.87506	.50337
45	$1 \cdot 00000$	$35\ 15\ 50$	.40446	.99554	.59108
50	$1 \cdot 19175$	$37\ 27\ 20$	.43360	$1 \cdot 12355$	.68995
55	$1 \cdot 42815$	$39\ 19\ 20$	.45913	$1 \cdot 26151$	.80238
60	$1 \cdot 73205$	$40\ 53\ 40$	.48117	$1 \cdot 41445$	.93328
65	$2 \cdot 14450$	$42\ 11\ 10$	.49971	$1 \cdot 58840$	$1 \cdot 08869$
70	$2 \cdot 74747$	$43\ 13\ 10$	.51478	$1 \cdot 79504$	$1 \cdot 28026$
75	$3 \cdot 73205$	$44\ 0\ 30$	.52646	$2 \cdot 05524$	$1 \cdot 52877$
80	$5 \cdot 67128$	$44\ 33\ 40$	.53469	$2 \cdot 41347$	$1 \cdot 87877$
85	$11 \cdot 43005$	$44\ 53\ 30$	.53973	$3 \cdot 02355$	$2 \cdot 48381$
90	∞	$45^\circ 0' 0''$	.54133	∞	∞

593.

A SHEEPSHANKS' PROBLEM (1866).

[From the *Messenger of Mathematics*, vol. iv. (1875), pp. 34—36.]

Apply the formulae of elliptic motion to determine the motion of a body let fall from the top of a tower at the equator.

The earth is regarded as rotating with the angular velocity ω round a fixed axis, so that the body is in fact projected from the apocentre with an angular velocity $= \omega$; and we write a for the equatorial radius, β for the height of the tower; then g denoting the force of gravity, and μ, h, n, a, e, θ , as in the theory of elliptic motion, we have

$$\begin{aligned} \mu &= n^2 a^3 = g a^2, \\ h &= (a + \beta)^2 \omega = n a^2 \sqrt{(1 - e^2)}, \\ a + \beta &= a(1 + e), \end{aligned}$$

whence

$$\begin{aligned}(a + \beta)^4 \omega^2 &= ga^3(1 - e^2), \\ (a + \beta) &= a(1 + e), \\ \frac{(a + \beta)^3 \omega^2}{ga^2} &= \frac{a\omega^2}{g} \left(1 + \frac{\beta}{a}\right)^3 = 1 - e,\end{aligned}$$

where $\frac{a\omega^2}{g}$ = ratio of centrifugal force to gravity,

$$= \frac{1}{289},$$

so that $1 - e$ is small,

$$r = \frac{a(1 - e^2)}{1 - e \cos \theta} = \frac{(a + \beta)(1 - e)}{1 - e \cos \theta}.$$

Suppose

$$\begin{aligned}r &= a, \\ 1 - e \cos \theta &= (1 - e) \left(1 + \frac{\beta}{a}\right) = 1 - e + \frac{\beta}{a}(1 - e),\end{aligned}$$

which is nearly

$$= 1 - e,$$

that is, θ is small, and therefore approximately

$$1 - e + \frac{1}{2}e\theta^2 = 1 - e + \frac{\beta}{a}(1 - e),$$

or

$$\theta^2 = \frac{2\beta}{a} \cdot \frac{1 - e}{e}.$$

We then have

$$\begin{aligned}r^2 d\theta &= h dt, \quad \text{or} \quad dt = \frac{r^2 d\theta}{h} = \frac{(a + \beta)^2(1 - e)^2}{(a + \beta)^2 \omega (1 - e \cos \theta)^2} d\theta \\ &= \frac{(1 - e)^2}{\omega(1 - e \cos \theta)^2} d\theta \\ &= \frac{(1 - e)^2}{\omega(1 - e + \frac{1}{2}e\theta^2)^2} d\theta \\ &= \frac{1}{\omega \left(1 + \frac{\frac{1}{2}e}{1 - e} \theta^2\right)^2} d\theta,\end{aligned}$$

that is,

$$\omega dt = \left(1 - \frac{e}{1 - e} \theta^2\right) d\theta.$$

Integrating, we have

$$\theta \left(1 - \frac{1}{3} \frac{e\theta^2}{1 - e}\right) = \omega t,$$

where ωt = earth's rotation in time t , = ϕ suppose; therefore

$$\theta \left(1 - \frac{1}{3} \frac{e\theta^2}{1 - e}\right) = \phi,$$

hence, if θ be as above, the angle described in falling to the surface,

$$\theta - \phi = \frac{1}{3} \frac{e\theta^3}{1 - e} = \frac{e\theta}{1 - e} \cdot \frac{\theta^2}{3} = \frac{e}{1 - e} \cdot \frac{1}{3} \cdot \frac{2\beta}{a} \theta.$$

Writing herein

$$\frac{1-e}{e} = \frac{a\omega^2}{g},$$

this is

$$\begin{aligned} &= \frac{2}{3} \frac{\beta}{a} \sqrt{\left(\frac{2\beta}{a} \cdot \frac{g}{a\omega^2}\right)} \sqrt{\left(\frac{a\omega^2}{g}\right)}, \\ &= \frac{2\beta}{3a} \cdot \frac{\beta}{a} \sqrt{\left(\frac{a\omega^2}{g}\right)}, \end{aligned}$$

viz.

$$\theta - \phi = \frac{2}{3a^2} \beta \sqrt{\left(\frac{2\beta}{a}\right)} \sqrt{\left(\frac{a\omega^2}{g}\right)},$$

whence

$$a(\theta - \phi) = \frac{2}{3} \sqrt{(2)} \sqrt{\left(\frac{\beta^3}{a}\right)} \sqrt{\left(\frac{a\omega^2}{g}\right)},$$

or say

$$\frac{a(\theta - \phi)}{\beta} = \frac{2\sqrt{(2)}}{3} \sqrt{\left(\frac{\beta}{a}\right)} \sqrt{\left(\frac{a\omega^2}{g}\right)},$$

where $a(\theta - \phi)$ is the distance at which the body falls from the foot of the tower. Substituting for $\sqrt{\left(\frac{a\omega^2}{g}\right)}$ its value, $= \frac{1}{17}$, we have

$$\frac{a(\theta - \phi)}{\beta} = \frac{2\sqrt{(2)}}{3 \cdot 17} \sqrt{\left(\frac{\beta}{a}\right)} = .056 \sqrt{\left(\frac{\beta}{a}\right)}.$$

594.

ON A DIFFERENTIAL EQUATION IN THE THEORY OF ELLIPTIC FUNCTIONS.

[From the *Messenger of Mathematics*, vol. iv. (1875), pp. 69, 70.]

The following equation presented itself to me in connexion with the cubic transformation:

$$Q^2 - Q\left(k + \frac{1}{k}\right) \cdot 3 = 3(1 - k^2) \frac{dQ}{dk}.$$

Writing as usual $k = u^4$, I was aware that a solution was

$$Q = \frac{v^2}{u^2} + 2uv,$$

where u, v are connected by the modular equation

$$u^4 - v^4 + 2uv(1 - u^2v^2) = 0;$$

but it was no easy matter to verify that the differential equation was satisfied. After a different solution, it occurred to me to obtain the relation between (Q, u) ; or, what is the same thing, (Q, k) , viz. eliminating v , we find

$$Q^2 - 6Q^2 - 4\left(u^4 + \frac{1}{u^4}\right)Q - 3 = 0,$$

or say

$$\frac{1}{Q}(Q^4 - 6Q^2 - 3) = 4\left(k + \frac{1}{k}\right);$$

whence also

$$\frac{1}{Q}(Q^2 \pm 8Q - 3) = 4\left(k \pm 2 + \frac{1}{k}\right);$$

that is,

$$\frac{1}{Q}(Q - 1)^2(Q + 3) = 4\left\{\sqrt{k} + \frac{1}{\sqrt{k}}\right\}^2,$$

and

$$\frac{1}{Q}(Q + 1)^2(Q - 3) = 4\left\{\sqrt{k} - \frac{1}{\sqrt{k}}\right\}^2,$$

and thence

$$\frac{(Q + 1)^2(Q - 3)}{(Q - 1)^2(Q + 3)} = \left(\frac{k - 1}{k + 1}\right)^2;$$

viz. the value of Q thus determined must satisfy the differential equation. This is easily verified, for, in virtue of the assumed integral, we have

$$Q^2 - 3 - \frac{1}{Q}(Q^4 - 6Q^2 - 3) = 3(1 - k^2)\frac{dQ}{dk};$$

that is,

$$Q^2 - 10Q^2 + 9 = -12(1 - k^2)\frac{dQ}{dk},$$

or finally

$$(Q^2 - 1)(Q^2 - 9) = -12(1 - k^2)\frac{dQ}{dk},$$

an equation which is at once obtained by differentiating logarithmically the former result, and we have thus the verification of the solution. This is, however, a particular integral only; and it appears doubtful whether there exists a general integral of an algebraical form.

595.

ON A SENATE-HOUSE PROBLEM.

[From the *Messenger of Mathematics*, vol. iv. (1875), pp. 75—78.]

The following was given [5 Jan., 1874,] as a problem of elementary algebra:

“Solve the equations

$$u(2a - x) = x(2a - y) = y(2a - z) = z(2a - u) = b^2,$$

and prove that unless $b^2 = 2a^2$, $x = y = z = u$, but that if $b^2 = 2a^2$, the equations are not independent.”

This is really a very remarkable theorem in regard to the intersections of a certain set of four quadric surfaces in four-dimensional space; viz. slightly altering the notation, we may write the equations in the form

$$x(2\theta - y) = m\theta^2 \dots (12),$$

$$y(2\theta - z) = m\theta^2 \dots (23),$$

$$z(2\theta - w) = m\theta^2 \dots (34),$$

$$w(2\theta - x) = m\theta^2 \dots (41),$$

where, regarding (x, y, z, w, θ) as coordinates in four-dimensional space, each equation represents a quadric surface. I remark that in such a space we have the notions, point-system, curve, subsurface, surface, according as the number of equations is 4, 3, 2, or 1.

Four quadric surfaces intersect in general in 16 points. But for the system in question (m being arbitrary), the common intersection consists of two lines and the two points

$$x = y = z = w = \theta\{1 \pm \sqrt{1 - m}\}$$

and in the case where $m = 2$, then the intersection consists of two lines and a certain unicursal quartic curve.

To obtain these results, I consider the four points

$$\begin{aligned}\theta = 0, \quad x = 0, \quad y = 0, \quad z = 0, \quad \dots 123, \\ \theta = 0, \quad y = 0, \quad z = 0, \quad w = 0, \quad \dots 234, \\ \theta = 0, \quad z = 0, \quad w = 0, \quad x = 0, \quad \dots 341, \\ \theta = 0, \quad w = 0, \quad x = 0, \quad y = 0, \quad \dots 412 : \end{aligned}$$

the two points

$$x = y = z = w = \theta\{1 \pm \sqrt{(1-m)}\}, \quad \dots PQ :$$

and the six lines

$$\begin{aligned}\theta = 0, \quad x = 0, \quad y = 0, \quad \dots 12, \\ \theta = 0, \quad y = 0, \quad z = 0, \quad \dots 23, \\ \theta = 0, \quad z = 0, \quad w = 0, \quad \dots 34, \\ \theta = 0, \quad w = 0, \quad x = 0, \quad \dots 41, \\ \theta = 0, \quad x = 0, \quad z = 0, \quad \dots 13, \\ \theta = 0, \quad y = 0, \quad w = 0, \quad \dots 24, \end{aligned}$$

being the edges of a tetrahedron, the vertices of which are the four points, viz. the point 123 is the intersection of the lines 12, 13, 23, and so for the other points.

The surfaces contain the several lines, viz.

$$\begin{aligned}\text{the surface 12 contains } & (12)^2, \quad 13, \quad 14, \quad 23, \quad 24, \\ & 23 \quad (23)^2, \quad 12, \quad 24, \quad 13, \quad 34, \\ & 34 \quad (34)^2, \quad 13, \quad 23, \quad 14, \quad 24, \\ & 41 \quad (41)^2, \quad 24, \quad 34, \quad 12, \quad 13, \end{aligned}$$

where $(12)^2$ denotes that 12 is a double line on the surface, and so in other cases. And it thus appears that the surfaces pass all four of them through the lines 13, 24, so that these lines are a part of the common intersection. To obtain the residual intersection, observe that the equations give

$$\begin{aligned}x = 2\theta - \frac{m\theta^2}{w} = \frac{m\theta^2}{2\theta - y}, \\ z = 2\theta - \frac{m\theta^2}{y} = \frac{m\theta^2}{2\theta - w}, \end{aligned}$$

whence

$$\begin{aligned}(2\theta - y)\left(2\theta - \frac{m\theta^2}{w}\right) = m\theta^2, \\ (2\theta - w)\left(2\theta - \frac{m\theta^2}{y}\right) = m\theta^2, \end{aligned}$$

or omitting from each equation the factor θ , the equations become

$$\begin{aligned}(2\theta - y)(2w - m\theta) = m\theta w, \\ (2\theta - w)(2y - m\theta) = m\theta y; \end{aligned}$$

that is,

$$\begin{aligned}(4 - 2m)\theta w - 2m\theta^2 - 2yw + m\theta(y + w) = 0, \\ (4 - 2m)\theta y - 2m\theta^2 - 2yw + m\theta(y + w) = 0. \end{aligned}$$

Whence, m not being $= 2$, we have $y = w$, and then

$$w^2 - 2\theta w + m\theta^2 = 0,$$

or, what is the same thing,

$$2\theta - w = \frac{m\theta^2}{w},$$

giving $x = y = z = w = \theta\{1 \pm \sqrt{(1-m)}\}$, viz. the surfaces each pass through the points P, Q . As regards the omitted factor θ , it is to be observed that, writing in the equations of the four surfaces $\theta = 0$, the equations become $xy = 0$, $yz = 0$, $zw = 0$, $wx = 0$, satisfied by $x = 0, z = 0$, or by $y = 0, w = 0$; we have thus $(\theta = 0, x = 0, z = 0)$ and $(\theta = 0, y = 0, w = 0)$, viz. the before-mentioned lines 13 and 24.

In the case $m = 2$, we have between y, w the single equation

$$yw - \theta(y + w) + 2\theta^2 = 0,$$

giving

$$y = \frac{\theta(w - 2\theta)}{w - \theta},$$

and thence

$$z = \frac{2\theta(w - \theta)}{w - 2\theta},$$

or, writing for convenience $a = \frac{w}{\theta}$, then the equations are

$$\begin{aligned} \frac{w}{\theta} &= a, \\ \frac{y}{\theta} &= \frac{a - 2}{a - 1}, \\ \frac{z}{\theta} &= \frac{-2}{a - 2}, \\ \frac{x}{\theta} &= \frac{2(a - 1)}{a}, \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} x &: 2(a - 1)^2(a - 2)\left(1 - \frac{a}{\infty}\right) \\ y &: a \dots (a - 2)^2\left(1 - \frac{a}{\infty}\right) \\ z &: -2a(a - 1) \dots \left(1 - \frac{a}{\infty}\right)^2 \\ w &: a^2(a - 1)(a - 2) \\ \theta &: a(a - 1)(a - 2)\left(1 - \frac{a}{\infty}\right), \end{aligned}$$

where, for the sake of homogeneity, I have introduced the factors $\left(1 - \frac{a}{\infty}\right)$ and $\left(1 - \frac{a}{\infty}\right)^2$, viz. we have x, y, z, w, θ proportional to quartic functions of the arbitrary parameter a , or the curve is a unicursal quartic. Writing in the equations $a = 0, 1, 2, \infty$ successively, we see that this quartic curve passes through the four points 123, 234, 341, 412 (intersecting at these points the lines 13 and 24 respectively); and writing also $a = 1 \pm i$ we see that the curve passes through the points P, Q , the coordinates of which now are

$$x = y = z = w = (1 \pm i)\theta.$$

It should admit of being proved by general considerations that, in 4-dimensional geometry when 4 quadric surfaces partially intersect in two lines, the residual intersection consists of 2 points; and that, when they intersect in the two lines and in a unicursal quartic met twice by each of the lines, there is no residual intersection—but this theory has not yet been developed.

596.

NOTE ON A THEOREM OF JACOBI'S FOR THE TRANSFORMATION OF A DOUBLE INTEGRAL.

[From the *Messenger of Mathematics*, vol. iv. (1875), pp. 92—94.]

Jacobi, in the Memoir “De Transformatione Integralis Duplicis...” &c., *Crelle*, t. VIII. (1832) pp. 253—279 and 321—357, [*Ges. Werke*, t. III., pp. 91—158], after establishing a theorem which includes the addition-theorem of elliptic functions, viz. this last is “the differential equation

$$\frac{d\eta}{\sqrt{(G''^2 \cos^2 \eta + G'''^2 \sin^2 \eta - G^2)}} + \frac{d\theta}{\sqrt{(G''^2 \cos^2 \theta + G'''^2 \sin^2 \theta - G^2)}} = 0,$$

has for its complete integral

$$G + G' \cos \eta \cos \theta + G'' \sin \eta \sin \theta = 0,$$

” (observe, as to the integral being complete, that the differential equation contains only the constant $G''^2 - G'''^2 \div (G'^2 - G''^2)$, whereas the integral equation contains the two constants $G' \div G$ and $G'' \div G$), obtains a corresponding theorem for double integrals; viz. this, in the corresponding special case, is as follows: If the variables (ϕ, ψ) and (η, θ) are connected by the two equations

$$\begin{aligned} \alpha \cos \phi + \alpha' \sin \phi \cos \psi \cdot \cos \eta \\ + \alpha'' \sin \phi \sin \psi \cdot \sin \eta \cos \theta \\ + \alpha''' \sin \phi \sin \psi \cdot \sin \eta \sin \theta = 0, \end{aligned}$$

$$\begin{aligned} \beta \cos \phi + \beta' \sin \phi \cos \psi \cdot \cos \eta \\ + \beta'' \sin \phi \cos \psi \cdot \sin \eta \cos \theta \\ + \beta''' \sin \phi \sin \psi \cdot \sin \eta \sin \theta = 0, \end{aligned}$$

and if putting for shortness

$$\begin{aligned} \alpha'' \beta''' - \alpha''' \beta'' &= f, & \alpha \beta' - \alpha' \beta &= a, \\ \alpha''' \beta' - \alpha' \beta''' &= g, & \alpha \beta'' - \alpha'' \beta &= b, \\ \alpha' \beta'' - \alpha'' \beta' &= h, & \alpha \beta''' - \alpha''' \beta &= c, \end{aligned}$$

(whence $af + bg + ch = 0$);

$$\begin{aligned} R^2 &= f^2 (\sin \phi \cos \psi)^2 (\sin \phi \sin \psi)^2 \\ &+ g^2 (\sin \phi \cos \psi)^2 (\cos \phi)^2 \\ &+ h^2 (\cos \phi)^2 (\sin \phi \cos \psi)^2 \\ &- a^2 (\cos \phi)^2 \\ &- b^2 (\sin \phi \cos \psi)^2 \\ &- c^2 (\sin \phi \sin \psi)^2, \end{aligned}$$

$$\begin{aligned} S^2 &= f^2 (\sin \eta \cos \theta)^2 (\sin \eta \sin \theta)^2 \\ &+ g^2 (\sin \eta \sin \theta)^2 (\cos \eta)^2 \\ &+ h^2 (\cos \eta)^2 (\sin \eta \cos \theta)^2 \\ &- a^2 (\cos \eta)^2 \\ &- b^2 (\sin \eta \cos \theta)^2 \\ &- c^2 (\sin \eta \sin \theta)^2, \end{aligned}$$

then we have

$$\frac{\sin \phi d\phi d\psi}{R} = \frac{\sin \eta d\eta d\theta}{S}.$$

And it may be added that the integral equations are, so to speak, a complete integral of the differential relation; viz. in virtue of the identity $af + bg + ch = 0$, the differential relation contains really only four constants; the integral relations contain the six constants $\alpha : \alpha' : \alpha'' : \alpha'''$ and $\beta : \beta' : \beta'' : \beta'''$, or we have two constants introduced by the integration.

The best form of statement is, in the first theorem, to write x, y for $\cos \eta, \sin \eta$, ($x^2 + y^2 = 1$), ξ, η for $\cos \theta, \sin \theta$, ($\xi^2 + \eta^2 = 1$), and similarly in the second theorem to introduce the variables x, y, z connected by $x^2 + y^2 + z^2 = 1$, and ξ, η, ζ connected by $\xi^2 + \eta^2 + \zeta^2 = 1$; then in the first theorem $d\eta, d\theta$ represent elements of circular arc, and in the second theorem $\sin \phi d\phi d\psi$ and $\sin \eta d\eta d\theta$ represent elements of spherical surface, and the theorems are:

I. If (x, y) are coordinates of a point on the circle $x^2 + y^2 = 1$, and (ξ, η) coordinates of a point on the circle $\xi^2 + \eta^2 = 1$, and if $ds, d\sigma$ are the corresponding circular elements, then

$$\frac{ds}{\sqrt{(ax^2 + by^2 - c)}} = \frac{d\sigma}{\sqrt{(a\xi^2 + b\eta^2 - c)}},$$

has for its complete integral

$$ax\xi + by\eta - c = 0.$$

II. If (x, y, z) are coordinates of a point on the sphere $x^2 + y^2 + z^2 = 1$, and (ξ, η, ζ) coordinates of a point on the sphere $\xi^2 + \eta^2 + \zeta^2 = 1$; and if $ds, d\sigma$ are the corresponding spherical elements, and

$$\begin{aligned} \beta\gamma' - \beta'\gamma &= f, & \alpha\delta' - \alpha'\delta &= a, \\ \gamma\alpha' - \gamma'\alpha &= g, & \beta\delta' - \beta'\delta &= b, \\ \alpha\beta' - \alpha'\beta &= h, & \gamma\delta' - \gamma'\delta &= c, \end{aligned}$$

(whence $af + bg + ch = 0$);

and for shortness

$$\begin{aligned} S^2 &= f^2 y^2 z^2 + g^2 z^2 x^2 + h^2 x^2 y^2 - a^2 x^2 - b^2 y^2 - c^2 z^2, \\ \Sigma^2 &= f^2 \eta^2 \zeta^2 + g^2 \zeta^2 \xi^2 + h^2 \xi^2 \eta^2 - a^2 \xi^2 - b^2 \eta^2 - c^2 \zeta^2, \end{aligned}$$

then the differential relation

$$\frac{ds}{\sqrt{(S)}} = \frac{d\sigma}{\sqrt{(\Sigma)}},$$

has for its complete integral the system

$$\begin{aligned} \alpha x\xi + \beta y\eta + \gamma z\zeta + \delta &= 0, \\ \alpha' x\xi + \beta' y\eta + \gamma' z\zeta + \delta' &= 0, \end{aligned}$$

where by complete integral is meant a system of two equations containing two arbitrary constants.

597.

ON A DIFFERENTIAL EQUATION IN THE THEORY OF ELLIPTIC FUNCTIONS.

[From the *Messenger of Mathematics*, vol. iv. (1875), pp. 110—113.]

The differential equation

$$Q^2 - Q\left(k + \frac{1}{k}\right) \cdot 3 = 3(1 - k^2) \frac{dQ}{dk},$$

considered ante, p. 69, [594, this volume, p. 244], belongs to a class of equations transformable into linear equations of the second order, and consequently is such that, knowing a particular solution, we can obtain the general solution.

In fact, assuming

$$Q = -3(1 - k^2) \frac{1}{z} \frac{dz}{dk},$$

the equation becomes

$$\begin{aligned} & 9(1-k^2)^2 \frac{1}{z^2} \left(\frac{dz}{dk} \right)^2 + 3(1-k^2) \left(k + \frac{1}{k} \right) \frac{1}{z} \frac{dz}{dk} - 3 \\ &= 3(1-k^2) \left\{ 3(1-k^2) \frac{1}{z^2} \left(\frac{dz}{dk} \right)^2 + 6k \frac{1}{z} \frac{dz}{dk} - 3(1-k^2) \frac{1}{z} \frac{d^2 z}{dk^2} \right\}, \end{aligned}$$

viz. omitting the terms in $\frac{1}{z^2} \left(\frac{dz}{dk} \right)^2$ which destroy each other, and dividing by $3(1-k^2)$, this is

$$\left(k + \frac{1}{k} \right) \frac{1}{z} \frac{dz}{dk} - \frac{1}{1-k^2} = 6k \frac{1}{z} \frac{dz}{dk} - 3(1-k^2) \frac{1}{z} \frac{d^2 z}{dk^2},$$

or finally

$$3(1-k^2) \frac{d^2 z}{dk^2} + \frac{1-5k^2}{k} \frac{dz}{dk} - \frac{1}{1-k^2} z = 0.$$

But knowing a particular value of Q we have

$$z = \exp \left\{ -\frac{1}{3} \int \frac{Q dk}{1-k^2} \right\},$$

a particular value of z , and thence in the ordinary manner the general value of z , giving the general value of Q .

The solution given in my former paper may be exhibited in a more simple form by introducing, instead of k , the variable a connected with it by the equation $k^2 = \frac{a^3(2+a)}{1+2a}$. We have in fact, *Fundamenta Nova*, p. 25, [*Jacobi's Ges. Werke*, t. I., p. 76],

$$v^2 = a \frac{2+a}{1+2a}, \quad u^2 = a^3 \frac{2+a}{1+2a},$$

viz. these expressions of u, v in terms of the parameter a , are equivalent to, and replace, the modular equation $u^4 - v^4 + 2uv(1 - u^2v^2) = 0$. We thence obtain

$$uv = a^2 \frac{2+a}{1+2a}, \quad \frac{v^2}{u^2} = \frac{1}{a^2} \left(\frac{2+a}{1+2a} \right),$$

that is,

$$uv = a^2 \left(\frac{2+a}{1+2a} \right), \quad \frac{v^2}{u^2} = \frac{1}{a^2} \left(\frac{2+a}{1+2a} \right),$$

and the particular solution, $Q = \frac{v^2}{u^2} + 2uv$, becomes

$$Q = \frac{1}{a^2} \sqrt{(1+2a)(2+a)} = \sqrt{\left\{ 5 + 2\left(a + \frac{1}{a}\right) \right\}}.$$

Introducing into the differential equation a in place of k , this is found to be

$$Q^2 - Q \frac{\frac{1}{a^2} + a^2 + 2\left(\frac{1}{a} + a\right)}{\sqrt{\left\{ 5 + 2\left(a + \frac{1}{a}\right) \right\}}} - 3 = (1-a^2) \sqrt{(1+2a)(2+a)} \frac{dQ}{da}.$$

But from this form it at once appears that it is convenient in place of a to introduce the new variable $\beta, = a + \frac{1}{a}$; the equation thus becomes

$$Q^2 - Q \frac{2-2\beta-\beta^2}{\sqrt{(5+2\beta)}} - 3 = (4-\beta^2) \sqrt{(5+2\beta)} \frac{dQ}{d\beta},$$

satisfied by $Q = \sqrt{(5 + 2\beta)}$; what or, what is the same thing, writing $5 + 2\beta = \gamma^2$, that is, $\beta = -\frac{5}{2} + \frac{1}{2}\gamma^2$, the equation becomes

$$4Q^2 + \frac{Q}{\gamma}(3 + 6\gamma^2 - \gamma^4) - 12 = -(\gamma^2 - 1)(\gamma^2 - 9) \frac{dQ}{d\gamma},$$

satisfied by $Q = \gamma$.

Writing here

$$Q = \frac{1}{2}(\gamma^2 - 1)(\gamma^2 - 9) \frac{1}{z} \frac{dz}{d\gamma},$$

we have for z the equation

$$(\gamma^2 - 1)(\gamma^2 - 9) \frac{d^2 z}{d\gamma^2} + (3\gamma^4 - 14\gamma^2 + 3) \frac{dz}{d\gamma} - \frac{48}{(\gamma^2 - 1)(\gamma^2 - 9)} z = 0,$$

satisfied by

$$z = \left(\frac{\gamma^2 - 9}{\gamma^2 - 1} \right)^{\frac{1}{2}}.$$

[In fact, this value gives

$$\frac{dz}{d\gamma} = 4\gamma(\gamma^2 - 9)^{-\frac{1}{2}}(\gamma^2 - 1)^{-\frac{3}{2}},$$

$$\frac{d^2 z}{d\gamma^2} = (-12\gamma^4 + 57\gamma^2 + 36)(\gamma^2 - 9)^{-\frac{3}{2}}(\gamma^2 - 1)^{-\frac{5}{2}},$$

which verify the equation as they should do.]

Representing for a moment the differential equation by $A \frac{d^2 z}{d\gamma^2} + B \frac{dz}{d\gamma} + Cz = 0$, and putting $z_1 = \left(\frac{\gamma^2 - 9}{\gamma^2 - 1} \right)^{\frac{1}{2}}$, then assuming $z = z_1 \int y d\gamma$, we find

$$A \left(z_1 \frac{dy}{d\gamma} + 2y \frac{dz_1}{d\gamma} \right) + By z_1 = 0,$$

that is,

$$\frac{1}{y} \frac{dy}{d\gamma} + \frac{2}{z_1} \frac{dz_1}{d\gamma} + \frac{B}{A} = 0,$$

viz.

$$\frac{1}{y} \frac{dy}{d\gamma} + \frac{2}{z_1} \frac{dz_1}{d\gamma} + \frac{3\gamma^4 - 14\gamma^2 + 3}{(\gamma^2 - 1)(\gamma^2 - 9)} = 0,$$

or

$$\frac{1}{y} \frac{dy}{d\gamma} + \frac{2}{z_1} \frac{dz_1}{d\gamma} + 3 + \frac{1}{\gamma^2 - 1} + \frac{15}{\gamma^2 - 9} = 0;$$

whence, integrating

$$\log y z_1^2 + 3\gamma - \frac{1}{2} \log \frac{\gamma + 1}{\gamma - 1} - \frac{15}{2} \log \frac{\gamma + 3}{\gamma - 3} = 0,$$

that is,

$$\begin{aligned} y &= e^{-3\gamma} \frac{1}{z_1^2} \left(\frac{\gamma + 1}{\gamma - 1} \right)^{\frac{1}{2}} \left(\frac{\gamma + 3}{\gamma - 3} \right)^{\frac{15}{2}} \\ &= e^{-3\gamma} \left(\frac{\gamma - 1}{\gamma - 3} \cdot \frac{\gamma + 3}{\gamma + 3} \right) \left(\frac{\gamma + 1}{\gamma - 1} \right)^{\frac{1}{2}} \left(\frac{\gamma + 3}{\gamma - 3} \right)^{\frac{1}{2}} \\ &= \frac{(\gamma + 1)(\gamma + 3)^2}{(\gamma - 3)^2} e^{-3\gamma}. \end{aligned}$$